

Formula

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} \cdot s_p$$

$$\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where s_p is the pooled standard deviation and $df = n_1$

$$+ n_2 - 2.$$

Example

Campaign data: $\bar{x}_1 - \bar{x}_2 = 4$; $s = 5.62$; $n_1=18$, $n_2=20$; 95%

$$CI \approx 4 \pm 2.028 \cdot 5.62 \cdot \sqrt{(1/18 + 1/20)} \Rightarrow (1.0, 6.0)$$

Interpretation

If 0 is outside the interval, the two-tailed test at $\alpha =$

0.05 would reject H_0 , indicating a significant difference between means.